

Problem Statement

Developers and founders waste hours manually scanning GitHub and Hacker News to find where to contribute, comment, or build reputation. EngageIQ automates this by ingesting 11,412 opportunities across 15 technical domains, scoring them via semantic embeddings and community signals, and adapting rankings in real time from user feedback.

System Architecture

Stage	Component	BAX-423 Lecture
1 · Ingestion	GitHub API · HN Firebase API Bloom Filter dedup · Python queue-based on-demand refresh HN titles resolved to real item IDs via Algolia search API (url_type: github_issue/github_repo/hn_item/hn_search_fallback)	Lecture 2
2 · Embedding	Sentence-BERT all-MiniLM-L6-v2 (384-dim) FAISS IndexFlatIP — exact search < 2 ms	Lecture 5
3 · Intent	7-intent classifier from role + interests Adaptive query expansion + candidate injection	app.py
4 · Scoring	Composite: 0.40 relevance + 0.30 community + 0.20 visibility + 0.10 (1–effort)	Lecture 7
5 · Ranking	Multi-stage: intent-aware rerank → diversity cap → Thompson Sampling bandit re-rank	Lecture 7
6 · Adaptation	Thompson Sampling (Beta-Bernoulli bandit) Domain preference learning over 50+ rounds	Lecture 8
7 · Analytics	Pandas batch: domain health, trending, volume	Lecture 3
8 · Dashboard	Streamlit: ranked cards, Why this?, CSV/JSON brief	—

Dataset

Metric	Value
Total records	11,412
Technical domains	15
GitHub records	8,588 (real API-derived issues + repos; direct links)
HN records	2,824 (690 resolved to real HN item URLs via Algolia multi-query match; 2,134 labeled hn_search_fallback, penalised −0.20; AQ top-5 ≤1 fallback cap)
Storage	CSV offline snapshot (committed to repo) + SQLite for live records
Live / offline split	11,412 offline snapshot · live records added via on-demand refresh

BAX-423 Techniques & Benchmarks

Integrated BAX-423 Techniques

Technique	Lecture	File	Role	Benchmark
Bloom Filter (sketching/dedup)	Lec 2	bloom_filter.py	Rejects duplicate streamed records before DB insertion	DB insertion rate on 11,412 unique IDs
Sentence-BERT + FAISS IndexFlatIP	Lec 5	embeddings.py	384-dim semantic embeddings; exact inner product search < 2 ms per query at 10k scale	Query latency: <2 ms Embedding NDCG@10: 1.00
Multi-stage Ranking (NDCG@10)	Lec 7	ranking.py	4-component composite score + diversity re-ranking NDCG@10 outperforms stars-only baseline	NDCG@10: 0.622 vs stars-only: 0.454
Thompson Sampling (RL bandit)	Lec 8	adaptive_learning.py	Beta-Bernoulli posterior per opportunity; domain preference score learns from feedback	50 rounds: ML/AI pref score 0.50→1.0

Ranking Benchmark — NDCG@10 (Sofia Persona, n=500)

Relevance defined as: 1.0 if domain in {Machine Learning, AI Research}, else 0.1. Sample: first 500 records. Random seed fixed at 42.

Ranking Method	NDCG@10	Notes
Random Baseline	0.4438	Shuffle — no signal
Stars-Only Ranking	0.4544	Popularity ≠ persona relevance
Embedding Similarity Only	1.0000	Semantic query match, no diversity
Full Composite + Re-rank ✓	0.6221	Balances relevance + community + diversity

Adaptive Learning — Thompson Sampling (50 Rounds)

Synthetic Sofia persona: engage if domain ∈ {ML, AI Research}, skip otherwise. Thompson Sampling Beta posterior updated each round. Random seed = 42.

Metric	Before (Round 0)	After (Round 50)
ML + AI Research domain pref score	0.50 (uniform prior)	1.00 (maximum)
Other 13 domains avg pref score	0.50 (uniform prior)	0.00 (minimized)
Preferred / non-preferred score gap	0.00	+1.00

6 Core Capabilities & Persona Test Results

6 Core Capabilities

#	Capability	Implementation
	Multi-Source Ingestion & On-Demand Refresh	GitHub API · Hacker News Firebase API Bloom Filter dedup · Python queue-based live refresh
2	Content Embedding & Similarity Retrieval	all-MiniLM-L6-v2 (384-dim) · FAISS IndexFlatIP Embedding cache for <2 ms queries
	Engagement Scoring & Multi-Stage Ranking	4-component composite + diversity re-rank NDCG@10 = 0.622 vs 0.444 random baseline
4	Adaptive Learning from Feedback	Thompson Sampling Beta-Bernoulli bandit Domain pref 0.50→1.00 over 50 rounds
	Batch Analytics & Trend Detection	Pandas batch: domain health, trending repos Volume-over-time, rising opportunities
6	Dashboard & Engagement Brief	Streamlit: ranked cards, Why this?, suggested actions CSV/JSON engagement brief export

Persona Pass/Fail Test Results

Each persona was tested against all 6 capabilities. Pass = feature surfaces expected results for that persona's interests.

Persona / Role	1 Ingest	2 Embed	3 Rank	4 Adapt	5 Analytics	6 Brief	Result
Sofia ML Student / Portfolio	✓ Pass	✓ Pass	✓ Pass	✓ Pass	✓ Pass	✓ Pass	PASS
David DevOps / Niche Community	✓ Pass	✓ Pass	✓ Pass	✓ Pass	✓ Pass	✓ Pass	PASS
Lina Data Journalist / Trends	✓ Pass	✓ Pass	✓ Pass	✓ Pass	✓ Pass	✓ Pass	PASS
Raj Startup Founder / B2B	✓ Pass	✓ Pass	✓ Pass	✓ Pass	✓ Pass	✓ Pass	PASS

Evidence notes: Sofia top-10 results surface ML Research and AI Research repos with semantic similarity ≥ 0.85 . David's results lead with DevOps/K8s and Cloud API repos. Lina's results include trending open-source across all domains sorted by growth_rate. Raj's results highlight Developer Tools and B2B SaaS repos by community engagement.

Hidden Persona Robustness — Adaptive Intent Layer (11/11 PASS)

The same intent inference, adaptive query expansion, and intent-aware reranking layer powers both the **main Action Queue** and the **Persona Test Panel**. Stress-tested against 11 roles not present in the system's ROLE_INTENT map. Intent is inferred from role + interest keywords; results evaluated against the same multi-condition pass criteria (domain_match ≥ 4 , primary_match ≥ 1 , src_fit ≥ 6 , neg=0).

Hidden Persona	Inferred Intent	domain	primary	src_fit	neg	Result
Security Researcher	security_review	5/10	5/10	10/10	0	■ PASS
Climate Tech Founder	startup_growth	7/10	4/10	10/10	0	■ PASS
Beginner Developer	contribution	5/10	4/10	10/10	0	■ PASS

Open Source Maintainer	community_engagement	8/10	4/10	10/10	0	■ PASS
Product Manager	startup_growth	7/10	7/10	10/10	0	■ PASS
Mobile Developer	mobile_contribution	8/10	8/10	10/10	0	■ PASS
Game Developer	generic	10/10	6/10	10/10	0	■ PASS
Data Engineer	data_engineering*	6/10	5/10	9/10	0	■ PASS
Academic ML Researcher	generic	10/10	8/10	10/10	0	■ PASS
Education Creator	trend_spotting	10/10	9/10	10/10	0	■ PASS
Privacy Researcher	security_review	5/10	5/10	10/10	0	■ PASS

* Data Engineer visible role maps to 'contribution' via ROLE_INTENT; hidden Data Engineer role triggers 'data_engineering' intent via keyword inference. Intent inference order: mobile > data_engineering > trend_spotting > contribution > startup_growth > security_review > community_engagement.

Limitations & Future Work

Honest Scope — Course-Project Implementation

Area	Current Implementation	Production Equivalent
Streaming	On-demand API refresh + Python queue/thread simulation	Apache Kafka / Flink persistent streaming cluster
Data Sources	GitHub + Hacker News (two sources meets spec requirements; Reddit excluded — OAuth2 API access unavailable in this scope)	Expand to authenticated APIs: Reddit PRAW, LinkedIn, DEV.to
AI Suggestions	Deterministic template-based engagement action generator (avoids API cost, latency, hallucination risk)	LLM-generated actions via Claude / GPT with prompt caching
Batch Analytics	Pandas batch over 11,412-record offline snapshot (sufficient for dataset size, <1 s query latency)	Apache Spark / Dask for distributed processing at scale
GH Archive	build_real_dataset.py includes GH Archive support; not used in runtime pipeline	Scheduled GH Archive pulls as supplemental source

Deployment Notes

App requires **sentence-transformers** and **faiss-cpu** (~450 MB combined). On first launch, embeddings are computed once and cached to `data/embeddings.npy` (~16 MB). Subsequent launches load the cache in <1 second. Streamlit Community Cloud deployment may require a free-tier instance with ≥1 GB RAM. Live API fetch is *optional* — the app runs fully on the pre-seeded offline dataset.

Future Work

- Connect suggested engagement actions to a live LLM (e.g., Claude claude-haiku-4-5-20251001) for natural-language engagement drafts; cache generated briefs to minimize API cost.
- Expand data sources to Reddit (authenticated PRAW), LinkedIn, and DEV.to for broader community coverage.
- Add scheduled GH Archive pulls for historical trend analysis beyond real-time API limits.
- Extend Thompson Sampling to opportunity-type preferences (issue vs. repo vs. post) for finer-grained adaptive ranking.
- Replace Pandas batch analytics with Dask or Spark when dataset grows beyond 100k records.

Submission Checklist

Item	Status
code/ — all .py files + requirements.txt	✓ Included
data/opportunities.csv — 11,412 offline records (GitHub + HN)	✓ Included
data/embeddings.npy — pre-computed 384-dim embeddings	✓ Included
brief.pdf — this document	✓ Included
prompts.md — development + planned runtime prompts	✓ Included
Live public URL	https://engageiq-bax423git-qianyingyang.streamlit.io
ZIP: Yang_Alice_BAX423_Final.zip	✓ Per one-pager spec